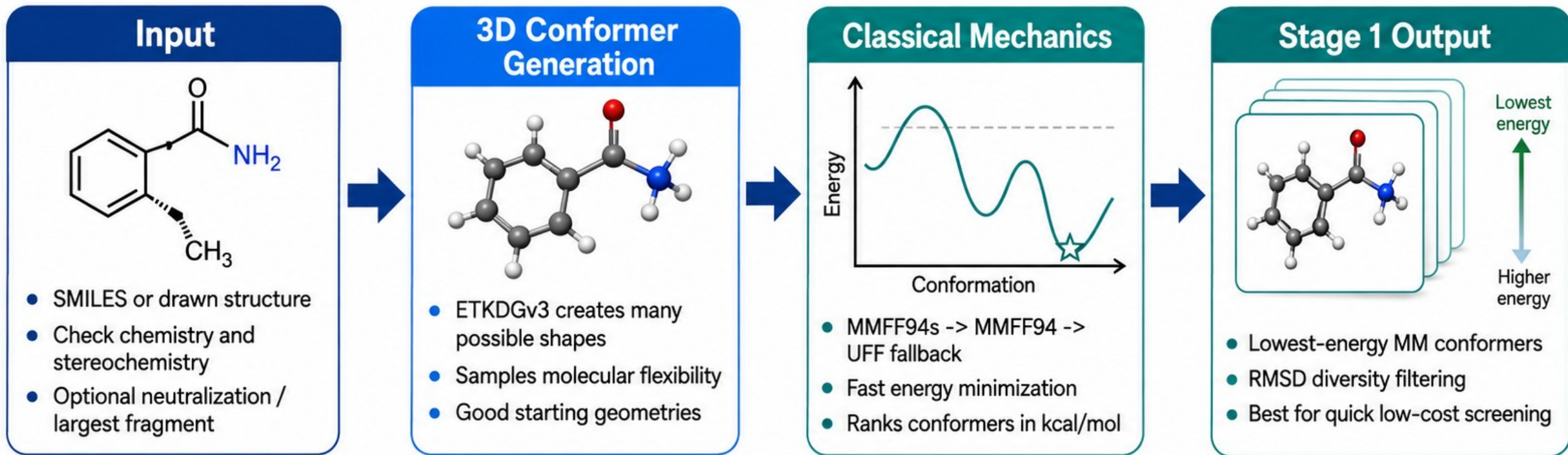


Molecular Optimizer Pro - Stage 1

Input, 3D conformers, and classical mechanics minimization



What it means

The program turns a 2D molecule into several 3D shapes, then quickly relaxes them to low-energy structures.



Why it matters

This is the fast first step that prepares realistic candidate geometries for deeper refinement.

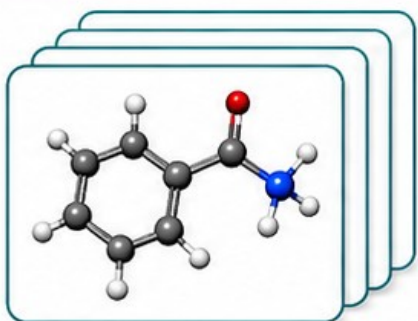


Stage 1 is fast and inexpensive - ideal for routine use and beginners.

Molecular Optimizer Pro - Stage 2

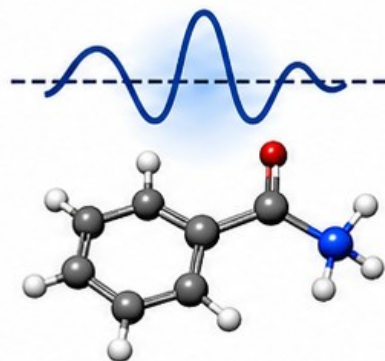
Semiempirical quantum refinement with GFN2-xTB

Stage 1 Input



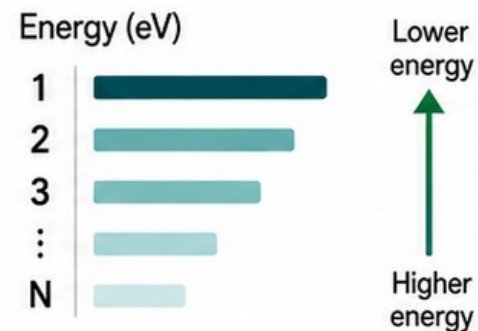
- Best MM conformers
- Energy-ranked and diversity filtered
- Starting geometries from Stage 1

GFN2-xTB Refinement



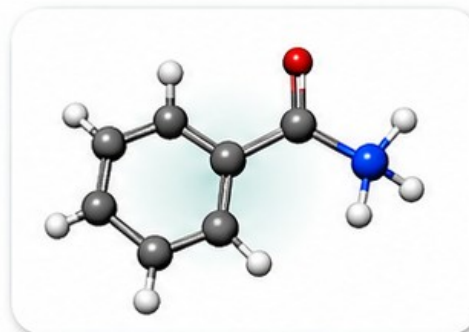
- Semiempirical quantum method
- Refines geometry more realistically
- Optimizes selected candidates

Energy Ranking



- Reports total energy in eV
- Compares candidates at the same method
- Selects the lowest xTB candidate

Stage 2 Output



- Improved low-energy geometry
- Better than MM for electronic effects
- Good balance of quality and cost



What it means

The program takes the best classical conformers and refines them using a faster quantum-mechanical approximation.



Why it matters

Stage 2 is a strong low-cost refinement step before expensive DFT or Gaussian calculations.

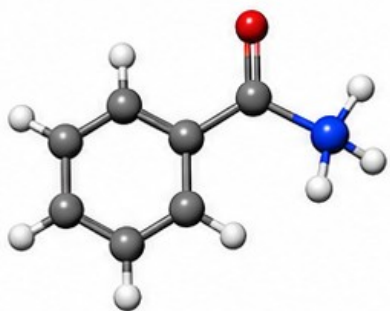


Stage 2 is moderately inexpensive - excellent for routine refinement before high-level quantum methods.

Molecular Optimizer Pro - Stage 3

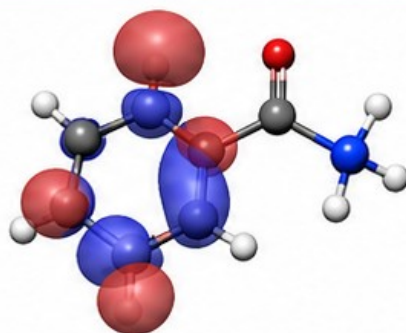
Optional DFT refinement in Colab

Input to DFT



- Best MM or xTB candidates
- User chooses DFT as an option
- Can include solvent setting

DFT Optimization



- PySCF density functional theory
- Improves geometry and electronic structure
- Uses more computer time

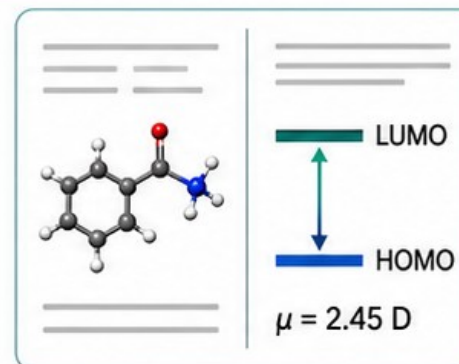
Verification

Progress



- ✓ Geometry optimization
 - ✓ Energy convergence
 - ✓ Frequency check (no significant imaginary frequencies)
-
- Tracks optimization progress
 - Optional frequency/Hessian check
 - Confirms a local minimum when no significant imaginary frequencies appear

Stage 3 Output



- Energy in Hartree
- HOMO-LUMO gap and dipole
- Most accurate Colab stage



What it means

This stage applies a higher-level quantum calculation to get a more rigorous molecular geometry and electronic description.



Why it matters

Use Stage 3 when you need better accuracy, but remember it is slower and more expensive than Stages 1 and 2.

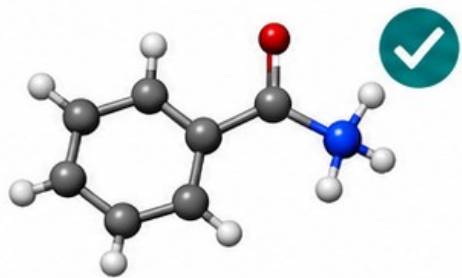


Stage 3 is accurate but expensive - best for selected important compounds.

Molecular Optimizer Pro - Stage 4

Gaussian / HPC export for high-level quantum mechanics

Starting Point



- Usually starts from Stage 2 xTB geometry
- Can also use a DFT-refined geometry
- Best candidate sent to HPC

Gaussian Package



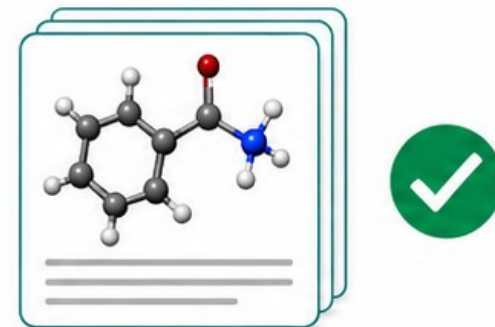
- Creates Gaussian input files
- Adds solvent and method settings
- Prepares restart and frequency jobs

HPC Submission



- Creates SLURM job scripts
- Recommends CPU, memory, and wall time
- Runs expensive optimization on a cluster

Stage 4 Output



- Optimized Gaussian geometry
- Optional frequency verification
- Professional HPC-ready workflow



What it means

The code prepares everything needed to move from Colab to a supercomputer for expensive Gaussian quantum calculations.



Why it matters

This stage is ideal when you need higher-level quantum optimization without spending Colab resources.



Stage 4 is for advanced HPC use - best for final high-level refinement and verification.